

Sriya Smita Sahoo

Chicago, IL | +1 (857) 339-8465 | sriyasahu.2003@gmail.com
[linkedin.com/in/sriya-smita-sahoo](https://www.linkedin.com/in/sriya-smita-sahoo) | github.com/ssahoo-lang | sriya-portfolio-neon.vercel.app

SUMMARY

Software engineer with production experience building backend services, full-stack web applications, and AI-enabled systems, and an M.S. in Computer Science. Ships end to end: relational schema and REST API design, event-driven workflow automation, LLM agent and retrieval-augmented generation (RAG) pipelines, and React/Next.js frontends. Strong in Python, FastAPI, PostgreSQL, and TypeScript, with a record of translating ambiguous operational requirements into maintainable, tested software that non-technical users actually adopt.

EDUCATION

Indiana University Bloomington

Bloomington, IN

Master of Science in Computer Science | GPA: 3.67/4.0

Aug. 2024 – May 2026

- *Coursework:* Applied Algorithms, Machine Learning, Computer Vision, Data Mining, Computer Networks, Distributed Systems, Cloud Computing

KL University

Vijayawada, India

Bachelor of Technology in Computer Science Engineering | GPA: 8.97/10

Aug. 2020 – May 2024

- *Coursework:* Database Management Systems, Design & Analysis of Algorithms, Operating Systems, Object-Oriented Programming, Cybersecurity, Edge Computing

EXPERIENCE

Heartland Community Network

Indiana, U.S. (Remote)

Senior Consultant

Jun. 2026 – Present

- Map workflow bottlenecks, system dependencies, and integration trade-offs across **5 business processes**, converting operational requirements into technical roadmaps for automation and data integration.
- Prototype automation and reporting solutions in Python and SQL to validate feasibility before engineering effort is committed, reducing downstream rework and scope churn.
- Drive requirements gathering, solution design, and technical documentation standards cross-functionally, aligning delivery priorities with business objectives, stakeholder expectations, and long-term system maintainability.

Indiana University Dining

Bloomington, IN

Inventory Data Analyst

Aug. 2025 – May 2026

- Engineered automated inventory-reporting and demand-trend pipelines across **5+ dining units**, converting raw operational data into near-real-time insight that reduced overstock and improved restocking decisions.
- Rebuilt ETL pipeline processes to eliminate inconsistent manual reporting across **3+ operational workflows**, standardizing documentation, data validation, and cross-functional logging practices.
- Designed SQL-backed reporting consumed directly by non-technical operators, prioritizing clarity and reliability over analytical complexity to drive adoption without training overhead.
- Surfaced recurring demand patterns through scheduled reporting that informed purchasing, replenishment, and operational planning, replacing ad-hoc spreadsheet requests with a repeatable reporting cycle.

Next Page Technologies

Hyderabad, India

Software Engineer

Jul. 2023 – Jul. 2024

- Owned **20+ modules** of a production CRM platform used daily by a live client team, designing relational schemas and RESTful backend services powering lead management and sales workflows.
- Delivered backend endpoints end to end, from data-model design and API contract through unit-tested implementation and deployment, translating business requirements into maintainable application logic.
- Automated event-driven workflows including rule-based processing, multi-step approval flows, notifications, and validation, standardizing execution across the full CRM lifecycle and reducing manual coordination between sales stages.
- Implemented role-based, stage-aware access control (RBAC) across frontend and backend layers, enforcing consistent authorization for users at different stages of the sales pipeline.
- Delivered releases in an Agile team through sprint planning, peer code review, and Git-based version control, resolving production defects reported directly by client users within active sprint cycles.

AICTE Edu Skills

Hyderabad, India

Robotic Process Automation Intern

Jan. 2022 – May 2022

- Cut manual data-processing time by **25%** across invoice and web-form workflows by building Python extraction and validation pipelines for repetitive operational tasks.
- Delivered **4 reusable automation solutions** by applying object-oriented design patterns to modularize RPA components and improve maintainability across workflows.
- Integrated REST-based email notification and report-generation services to surface automation outcomes, validation results, and exception cases to stakeholders without manual follow-up.
- Translated process requirements into technical designs alongside cross-functional stakeholders, validating workflow logic, exception paths, and edge-case handling with process owners ahead of production rollout.

PROJECTS

- NoteKit – Grounded RAG Learning Platform** | *Python, FastAPI, PostgreSQL, pgvector, Next.js* | [GitHub](#) Aug. 2026
- Built an end-to-end retrieval-augmented generation (RAG) platform that turns web sources and user uploads into citation-backed study modules and auto-generated practice quizzes.
 - Achieved **95.3% claim faithfulness** and **16/16 correct refusals** on a fixed evaluation benchmark by combining dense-vector retrieval (pgvector) with PostgreSQL full-text search through reciprocal-rank fusion and cross-encoder reranking.
 - Reduced first-response latency by **65% (52s to 18s)** and quiz-enabled course cost by **48%** through concurrent **asyncio** processing, server-sent events (SSE) streaming, and prompt caching.
 - Designed ingestion, chunking, embedding, and retrieval as independently testable services to support evaluation-driven iteration on retrieval quality.
- RatePilot – Autonomous Bill Negotiation Agent** | *Python, FastAPI, React, LLM APIs, Gmail API* | [GitHub](#) May 2026
- Built a full-stack autonomous LLM agent that researches live competitor pricing through a web-search tool, formulates negotiation strategy, and generates personalized outreach with **no manual input after upload**.
 - Designed stateful accept, counter, and escalate decision logic that selects each next action from negotiation context, prior correspondence, and retrieved pricing evidence, eliminating reliance on unsupported model assumptions.
 - Integrated Gmail API monitoring for inbound replies and async PDF parsing with structured LLM extraction, backed by a React dashboard surfacing per-negotiation and monthly savings outcomes.
 - Maintained a complete cross-session audit trail of agent decisions, rationale, and state transitions, making autonomous behavior inspectable and debuggable.
- Distributed Graph Processing Engine** | *Python, Apache Spark, GraphX, Scala, Docker, Distributed Systems* [GitHub](#) Apr. 2026
- Automated **300+ benchmark runs** across PageRank and SSSP algorithms on **5M-edge** Stanford SNAP datasets by deploying a distributed engine on a 5-node NSF Jetstream2 cluster with variable worker and partitioning configurations.
 - Achieved a **3.3× speedup** on large-graph workloads and quantified **25–400×** Spark overhead on sub-RAM datasets by profiling task-scheduling bottlenecks (GC <2.5%, shuffle-wait <2.7%).
 - Reduced cloud deployment wall-time by **31.7%** by tuning Spark data-locality parameters, establishing a data-driven threshold for distributed vs. single-node deployment decisions.
- Real-Time Edge AI Vision System** | *Python, YOLOv8, Raspberry Pi, OpenCV* | [GitHub](#) Sep. 2025
- Improved on-device inference throughput by **40%** with zero cloud dependency by building an edge-deployed vehicle-detection and speed-monitoring system on Raspberry Pi using YOLOv8, OpenCV, and multi-object tracking.
 - Delivered **sub-second inference** on constrained hardware by optimizing frame-analysis workflows and integrating automated violation detection, alerting, and centralized logging.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C, C++, Scala, HTML/CSS, Bash
Backend: FastAPI, Flask, Django, Node.js, REST APIs, Microservices, Async/Concurrency, WebSockets, Pydantic, SQLAlchemy
Frontend: React.js, Next.js, Vite, Responsive Design, Component Architecture, State Management, Server-Sent Events (SSE)
AI/ML: LLM Applications, Agentic AI, Tool Use, Prompt Engineering, RAG, Vector Embeddings, Semantic Search, LangChain, LangGraph, Transformers, YOLOv8, OpenCV, PyTorch, TensorFlow, scikit-learn, pandas, NumPy
Data & Databases: PostgreSQL, pgvector, MySQL, SQLite, Schema Design, Query Optimization, Indexing, Full-Text Search, ETL & Data Pipelines, Apache Spark
Cloud & DevOps: AWS (EC2, S3), Azure, GCP, Docker, CI/CD, Git/GitHub, Linux/Unix, Monitoring & Logging
Practices: System Design, Object-Oriented Design, Data Structures & Algorithms, API Integration, Unit Testing (pytest), Code Review, Agile/Scrum, Authentication & Authorization (OAuth2, JWT, RBAC), Technical Documentation

CERTIFICATIONS & PUBLICATION

Certifications: AWS Certified Solutions Architect – Associate; AWS Certified Cloud Practitioner (CLF-C02); Microsoft Azure Fundamentals (AZ-900); Red Hat Certified Enterprise Application Developer; Google Cloud Associate Cloud Engineer.
Publication: “Survey of Non-English Language Compilers: Exploring the Diversity of Programming Languages,” *2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India, pp. 1187–1193.